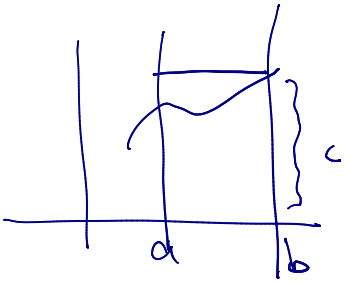
8.8 Improper Integrals

We have considered definite integrals of the form $I = \int_a^b f(x)dx$,

where the integrand f is ***continuous*** on the ***closed, finite*** interval $[a, b]$.

The integral I is necessarily a finite number. Such integrals are also called ***proper integrals***.

Definite integral: $I = \int_a^b f(x)dx$ → finite if $f(x)$ is continuous in $[a, b]$.



- **Proper**
  f is continuous on the closed, finite interval $[a, b]$.
- **Improper** $\int_0^\infty \frac{1}{1+x^2}dx$
  - Type I: $a \to -\infty$ or $b \to \infty$. *(Domain is infinite)*
  - Type II: $f(x) \to \pm\infty$ at some points on $[a, b]$. *(Range is infinite)*

$\int_0^2 \frac{1}{1-x^2}dx$ → is not defined when $x = 1$

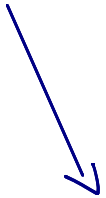
Example:

$\int_0^2 \frac{1}{1+x^2}\,dx$ Proper as $\frac{1}{1+x^2}$ is continuous on $[0,2]$.

$\int_1^\infty \frac{1}{x(1+x^2)}\,dx$ Improper, Type 1 as domain is infinite.

$\int_0^2 \frac{1}{1-x^2}\,dx$ Improper, Type 2 as $\frac{1}{1-x^2}$ is not defined @ $x=1$.

$\int_0^1 \frac{1}{1-x^2}\,dx + \int_1^2 \frac{1}{1-x^2}\,dx$



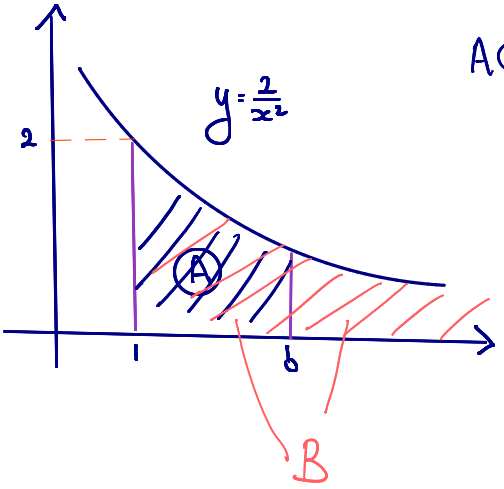
do not integrate directly

Remark:

If $f(x) \geq 0$ over an interval $[a, b]$, the area under the curve $y = f(x)$ is given by $\int_a^b f(x)dx$.

An improper integral (Type I or Type II) can also be interpreted as an area. The area is infinite if the improper integral diverges.

Example: $\int_1^\infty \frac{2}{x^2}\,dx$



$A(b) = \int_1^b \frac{2}{x^2}\,dx$

$\int_1^\infty \frac{2}{x^2}\,dx = \lim_{b\to\infty} A(b)$

$B = \lim_{b\to\infty} \int_1^b \frac{2}{x^2}\,dx$

Type I Improper Integrals

Integrals with infinite limits of integration are **improper integrals of Type I.**

1. If $f(x)$ is continuous on $[a, \infty)$, then

$$\int_a^{\infty} f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx.$$

treat as a function of b

2. If $f(x)$ is continuous on $(-\infty, b]$, then

$$\int_{-\infty}^{b} f(x)\,dx = \lim_{a\to-\infty} \int_a^b f(x)\,dx.$$

treat as a fn of a

3. If $f(x)$ is continuous on $(-\infty, \infty)$, then

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{-\infty}^{c} f(x)\,dx + \int_c^{\infty} f(x)\,dx,$$

where c is any real number.

$= \lim_{a\to-\infty} \int_a^c f(x)\,dx$ $\qquad$ $\lim_{b\to\infty} \int_c^b f(x)\,dx$.

In each case, if the limit is finite we say that the improper integral **converges** and that the limit is the **value** of the improper integral. If the limit fails to exist, the improper integral **diverges**.

Example 1

Is the area under the curve $y = (\ln x)/x^2$ from $x = 1$ to $x = \infty$ finite? If so, what is it?

$$\text{Area} : \int_1^\infty \frac{\ln x}{x^2}\,dx = \lim_{b\to\infty} \int_1^b \frac{\ln x}{x^2}\,dx$$

$u = \ln x \Rightarrow du = \frac{1}{x}\,dx$

$dv = \frac{1}{x^2}\,dx \Rightarrow v = -\frac{1}{x}$

$$= \lim_{b\to\infty} \left(\left[-\frac{\ln x}{x} \right]_1^b - \int_1^b \left(-\frac{1}{x}\right)\left(\frac{1}{x}\right) dx \right)$$

$$= \lim_{b\to\infty} \left(-\frac{\ln b}{b} - \left[\frac{1}{x} \right]_1^b \right) = \lim_{b\to\infty} \left(-\frac{\ln b}{b} - \frac{1}{b} + 1 \right)$$

$\frac{\ln b}{b} \to 0$, $\frac{1}{b} \to 0$ (L'Hopital)

$$= 1 - \lim_{b\to\infty} \frac{1/b}{b} = 1$$

The area converges to the value 1.

$$\ln x \ll x \ll 4x^2 + 5$$

eg. $(\ln x)^{10000000000}$

$\ll$

$x^{0.000000000001}$

log

$\ll$

polynomial

$\ll$

exponential

Example 2

Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$.

$= \int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + \int_{0}^{\infty} \frac{1}{1+x^2}\,dx$ (A + B)

$A = \lim_{a \to -\infty} \int_a^0 \frac{1}{1+x^2}\,dx = \lim_{a \to -\infty} \left[\tan^{-1}(x)\right]_a^0$

$= \lim_{a \to -\infty} \left(\tan^{-1}(0) - \tan^{-1}(a)\right)$ ($\tan^{-1}(0) \to 0$; $-\tan^{-1}(a) \to -\frac{\pi}{2}$ as $a \to -\infty$)

$= -\left(-\frac{\pi}{2}\right) = \frac{\pi}{2}$

Similarly, $B = \frac{\pi}{2}$

$\therefore \int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \frac{\pi}{2} + \frac{\pi}{2} = \pi$

The integral converges to the value π.

Example 3

For what values of p does the integral $\int_1^\infty \frac{1}{x^p}\,dx$ converge? When the integral does converge, what is its value?

For $p \neq 1$, we have $\int_1^\infty \frac{1}{x^p}\,dx = \lim_{b\to\infty} \int_1^b \frac{1}{x^p}\,dx$

$$= \lim_{b\to\infty} \left[\frac{x^{-p+1}}{-p+1}\right]_1^b$$

$$= \frac{1}{1-p} \lim_{b\to\infty} \left(\frac{1}{b^{p-1}} - 1\right)$$

$$= \begin{cases} \frac{1}{p-1} & \text{if } p > 1 \quad \left(\because \lim_{b\to\infty} \frac{1}{b^{p-1}} = 0 \text{ if } p>1\right) \\ \to \infty & \text{if } p < 1 \quad \left(\lim_{b\to\infty} \frac{1}{b^{p-1}} \to \infty \text{ if } p<1\right) \end{cases}$$

For $p = 1$, $\int_1^\infty \frac{1}{x}\,dx = \lim_{b\to\infty} \int_1^b \frac{1}{x}\,dx$

$$= \lim_{b\to\infty} [\ln x]_1^b = \lim_{b\to\infty} (\ln b) = \infty$$

$$\therefore \int_1^\infty \frac{1}{x^p}\,dx = \begin{cases} \frac{1}{p-1} & \text{if } p > 1 \quad (\text{converges}) \\ \infty & \text{if } p \leq 1 \quad (\text{diverges}) \end{cases}$$

Type II Improper Integrals: Integrals of functions that become infinite at a point within the interval of integration are **improper integrals of Type II.**

(Range infinite)

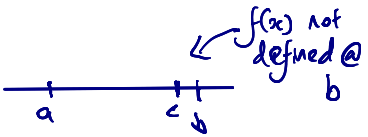
1. If $f(x)$ is continuous on $(a, b]$ and discontinuous at a, then



$$\int_a^b f(x)\,dx = \lim_{c\to a^+} \boxed{\int_c^b f(x)\,dx.}$$

treat as a function of c

2. If $f(x)$ is continuous on $[a, b)$ and discontinuous at b, then



$$\int_a^b f(x)\,dx = \lim_{c\to b^-} \boxed{\int_a^c f(x)\,dx.}$$

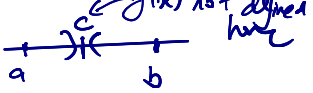
treat as a fn of c.

3. If $f(x)$ is discontinuous at c, where $a < c < b$, and continuous on $[a, c) \cup (c, b]$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

$= \lim_{d\to c^-} \int_a^d f(x)\,dx + \lim_{e\to c^+} \int_e^b f(x)\,dx.$

where c is any real number.



In each case, if the limit is finite we say that the improper integral **converges** and that the limit is the **value** of the improper integral. If the limit fails to exist, the improper integral **diverges**.

Example 4

Investigate the convergence of

$$\int_0^1 \frac{1}{1-x}\,dx.$$

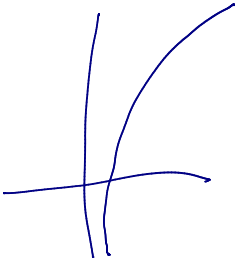
discontinuous @ $x = 1$

$$= \lim_{b \to 1^-} \int_0^b \frac{1}{1-x}\,dx$$

$$= \lim_{b \to 1^-} \left[-\ln|1-x| \right]_0^b = \lim_{b \to 1^-} -\ln|1-b|$$

$$\longrightarrow \infty$$

The integral diverges.



Example 5

Evaluate $\int_0^3 \frac{dx}{(x-1)^{2/3}}$.

$\frac{1}{(x-1)^{2/3}}$ is discontinuous @ $x = 1$

$$= \underbrace{\int_0^1 \frac{dx}{(x-1)^{2/3}}}_{A} + \underbrace{\int_1^3 \frac{dx}{(x-1)^{2/3}}}_{B}$$

$$A = \lim_{b \to 1^-} \int_0^b \frac{dx}{(x-1)^{2/3}} = \lim_{b \to 1^-} \left[3(x-1)^{1/3}\right]_0^b$$

$$= \lim_{b \to 1^-} \left[3(b-1)^{1/3} - 3(-1)^{1/3}\right]$$

$$= 3$$

$$B = \lim_{b \to 1^+} \int_b^3 \frac{dx}{(x-1)^{2/3}} = \lim_{b \to 1^+} \left[3(x-1)^{1/3}\right]_b^3 = \lim_{b \to 1^+} \left[3 \cdot 2^{1/3} - 3(b-1)^{1/3}\right]$$

$$= 3 \cdot 2^{1/3}$$

$$\Rightarrow \int_0^3 \frac{dx}{(x-1)^{2/3}} = A + B = \underline{3 + 3 \cdot 2^{1/3}} \quad \text{(converges)}$$

Example 6

discontinuous @ $x = 1$, but $1 \notin [2, \infty)$

Type 1 Improper.

$$\int_2^{\infty} \frac{x+3}{(x-1)(x^2+1)}\,dx = \ln 5 + \tan^{-1}(2) - \frac{\pi}{2} \approx 1.1458.$$

partial fractions

$$= \int_2^{\infty} \frac{2}{x-1} - \frac{2x+1}{x^2+1}\,dx = \lim_{b\to\infty} \int_2^b \left(\frac{2}{x-1} - \frac{2x}{x^2+1} - \frac{1}{x^2+1}\right)dx$$

$$= \lim_{b\to\infty} \left[2\ln|x-1| - \ln(x^2+1) - \tan^{-1}(x)\right]_2^b$$

$$= \lim_{b\to\infty} \left(2\ln(b-1) - \ln(b^2+1) - \tan^{-1}(b) + \ln 5 + \tan^{-1} 2\right)$$

$\tan^{-1}(b) \to \frac{\pi}{2}$

$$= \ln 5 + \tan^{-1} 2 - \frac{\pi}{2} + \lim_{b\to\infty} \ln\left[\frac{(b-1)^2}{b^2+1}\right]$$

$$= \ln 5 + \tan^{-1} 2 - \frac{\pi}{2} + \ln\left[\lim_{b\to\infty} \left(\frac{b^2-2b+1}{b^2+1}\right)\right]$$

l'Hôpital

$$\lim_{b\to\infty} \frac{2b-2}{2b} = \lim_{b\to\infty} \left(1 - \frac{1}{b}\right) = 1$$

$\ln 1 \to 0$

$$= \ln 5 + \tan^{-1} 2 - \frac{\pi}{2} + \ln 1$$

$$= \ln 5 + \tan^{-1} 2 - \frac{\pi}{2}$$

Caution: It is **incorrect** to write

- $\displaystyle\int_{-1}^{2} \frac{1}{x^2}dx = \left[-\frac{1}{x}\right]_{-1}^{2} = -\frac{3}{2}.$ ✗

  ↳ discontinuous @ $x=0$

  $\displaystyle\int_{-1}^{0} \frac{1}{x^2}dx + \int_{0}^{2} \frac{1}{x^2}dx$ → diverges

  $\lim_{b\to 0^-} \int_{-1}^{b} \frac{1}{x^2}dx = \lim_{b\to 0^-} \left[-\frac{1}{x}\right]_{-1}^{b}$

  $= \lim_{b\to 0^-} \left(1 - \frac{1}{b}\right) \to \infty$

  does not converge.

- $\displaystyle\int_{0}^{4} \frac{1}{\sqrt{x}}dx = \left[2\sqrt{x}\right]_{0}^{4} = 4.$

  ↳ $\lim_{a\to 0^+} \int_{a}^{4} \frac{1}{\sqrt{x}}dx = \ldots = 4$

Do not apply FTC when $f(x)$ is discontinuous on $[a, b]$.

The two examples above are **Type II Improper Integrals.**

★ When an integral with finite limits of integration are given, check whether it is proper or improper.

Example 7

Evaluate $\int_0^3 \frac{dx}{x-1}$.

Wrong working: $\int_0^3 \frac{dx}{x-1} = \left[\ln|x-1|\right]_0^3 = \ln 2 - \ln 1 = \ln 2$ ✗

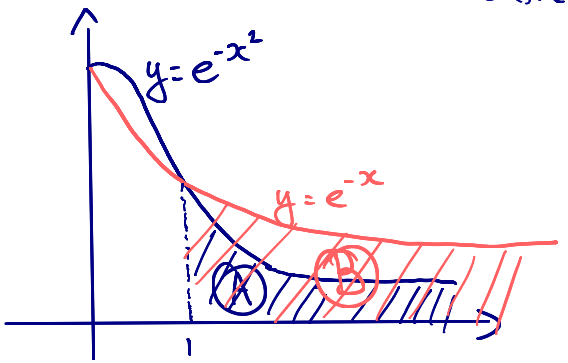
$$\int_0^3 \frac{dx}{x-1} = \int_0^1 \frac{1}{x-1}\,dx + \int_1^3 \frac{1}{x-1}\,dx$$

diverges

$\Rightarrow \int_0^3 \frac{dx}{x-1}$ diverges

Example 8

Does the integral $\int_1^\infty e^{-x^2}\,dx$ converge?



Observe that $0 \le e^{-x^2} \le e^{-x}$ for $x \ge 1$.

$\int_1^\infty e^{-x^2}\,dx$ = Area (A)

$\int_1^\infty e^{-x}\,dx$ = Area (B) $= e^{-1}$

Area (B) $\ge$ Area (A)

$$\Rightarrow \int_1^\infty e^{-x^2}\,dx \le \int_1^\infty e^{-x}\,dx = \lim_{b\to\infty}\left[-e^{-x}\right]_1^b = \lim_{b\to\infty}\left(e^{-1} - e^{-b}\right)$$

($e^{-b} \to 0$)

$$= e^{-1}$$

$\Rightarrow$ $\int_1^\infty e^{-x^2}\,dx$ converges to a value $\le e^{-1}$.

What if we ask for $\int_0^\infty e^{-x^2}\,dx = \int_0^1 e^{-x^2}\,dx + \int_1^\infty e^{-x^2}\,dx$?

$\int_0^1 e^{-x^2}\,dx$: bounded region, finite, converges

$\int_1^\infty e^{-x^2}\,dx$: converges

$\therefore \int_0^\infty e^{-x^2}\,dx$ converges.

Tests for Convergence and Divergence

When an improper integral cannot be evaluated, we may still be able to determine whether the integral converges (or diverges) by comparing it with one integral whose convergence (divergence) properties are known.

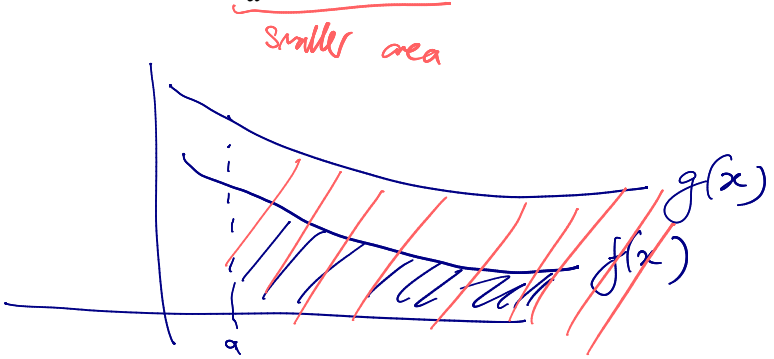
$\int_a^\infty f(x)\,dx$ has a "smaller" area compared to $\int_a^\infty g(x)\,dx$.

Theorem: Direct Comparison Test

Let f and g be continuous on $[a, \infty)$ with $0 \le f(x) \le g(x)$ for all $x \ge a$. Then

1. $\int_a^\infty f(x)\,dx$ converges if $\int_a^\infty g(x)\,dx$ converges

2. $\int_a^\infty g(x)\,dx$ diverges if $\int_a^\infty f(x)\,dx$ diverges.



Example 9

- $\int_1^\infty \frac{\sin^2 x}{x^2} dx$ converges

$0 \le \sin^2 x \le 1 \Rightarrow 0 \le \frac{\sin^2 x}{x^2} \le \frac{1}{x^2}$

$\Rightarrow \int_1^\infty \frac{\sin^2 x}{x^2} dx \le \boxed{\int_1^\infty \frac{1}{x^2} dx}$ converges

$\Rightarrow \int_1^\infty \frac{\sin^2 x}{x^2} dx$ converges.

- $\int_1^\infty \frac{1}{\sqrt{x^2-0.1}} dx$ diverges

$\int_1^\infty \frac{1}{x} dx$ diverges so

$\frac{1}{\sqrt{x^2-1}} > \frac{1}{\sqrt{x^2}}$ for $x \ge 1$

$= \frac{1}{x}$

by DCT, $\int_1^\infty \frac{1}{\sqrt{x^2-0.1}} dx$ also diverges.

Caution: Do Not Use the Direct Comparison Test to test $\int_a^\infty f(x)\,dx$ when

i. $f(x) \le g(x)$ and $\int_a^\infty g(x)\,dx$ diverges; or

The larger area $\to \infty$, so we know nothing about the smaller area.

ii. $g(x) \le f(x)$ and $\int_a^\infty g(x)\,dx$ converges.

The small area converges, so we know nothing about the larger area.

For example, to determine if $\int_1^\infty \frac{1-e^{-x}}{x}dx$ converges, we cannot compare it with $\int_1^\infty \frac{1}{x}dx$ even though we know $\frac{1-e^{-x}}{x} \le \frac{1}{x}$. This is because $\int_1^\infty \frac{1}{x}dx$ diverges, and we cannot say anything about $\int_1^\infty \frac{1-e^{-x}}{x}dx$. (In particular, we cannot deduce from direct comparison test that it diverges.) We use limit comparison test instead.

smaller area

larger area

Theorem: Limit Comparison Test

If the positive functions f and g are continuous on $[a, \infty)$ and if

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L,\ 0 < L < \infty,$$

then $\int_a^\infty f(x)\,dx$ and $\int_a^\infty g(x)$ both converge or both diverge.

$\int_1^\infty \frac{1}{x^p}\,dx$

converges when $p > 1$

diverges when $p \le 1$.

Example: Test $\int_1^\infty \frac{1-e^{-x}}{x}\,dx$ for convergence.

$f(x) = \frac{1-e^{-x}}{x}$ $\quad$ $g(x) = \frac{1}{x}$

$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{1-e^{-x}}{1} = 1$

LCT says that $\int_1^\infty f(x)\,dx$ converges $\iff$ $\int_1^\infty g(x)\,dx$ converges.

Since $\int_1^\infty \frac{1}{x}\,dx$ diverges, so $\int_1^\infty \frac{1-e^{-x}}{x}\,dx$ also diverges.

Example 10 $f(x) = \frac{1}{1+x^2}$ $g(x) = \frac{1}{x^2}$

Show that $\int_1^\infty \frac{dx}{1+x^2}$ converges by comparison with $\int_1^\infty \frac{1}{x^2}\,dx$. Find and compare the integral values.

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{1/1+x^2}{1/x^2} = \lim_{x\to\infty} \frac{x^2}{1+x^2} = \lim_{x\to\infty} \frac{1}{1+\frac{1}{x^2}} = 1$$

We know $\int_1^\infty \frac{1}{x^2}\,dx$ converges, so by LCT $\int_1^\infty \frac{dx}{1+x^2}$ converges too.

But $\int_1^\infty \frac{1}{1+x^2}\,dx$ is not necessarily equal to $\int_1^\infty \frac{1}{x^2}\,dx$.

- $\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty} \int_1^b \frac{1}{x^2}\,dx = \lim_{b\to\infty} \left[-\frac{1}{x}\right]_1^b = \lim_{b\to\infty} \left(1-\frac{1}{b}\right) = 1$
- $\int_1^\infty \frac{1}{1+x^2}\,dx = \lim_{b\to\infty} \left[\tan^{-1}x\right]_1^b = \lim_{b\to\infty} \left(\tan^{-1}b - \tan^{-1}1\right) = \frac{\pi}{4}$.

Here $\int_1^\infty \frac{1}{1+x^2}\,dx \neq \int_1^\infty \frac{1}{x^2}\,dx$.

Example 11

Show that $\int_1^{\infty} \underbrace{\frac{3}{e^x+5}}_{f(x)}\,dx$ converges. $g(x) = \frac{3}{e^x}$ (OR $\frac{1}{e^x}$)

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = \lim_{x\to\infty} \frac{e^x}{e^x+5} = \lim_{x\to\infty} \frac{1}{1+5/e^x} = 1$$

By LCT, $\int_1^{\infty} \frac{3}{e^x+5}\,dx$ converges $\Leftrightarrow$ $\int_1^{\infty} \frac{3}{e^x}\,dx$ converges.

$$\int_1^{\infty} \frac{3}{e^x}\,dx = \lim_{b\to\infty} \left[-\frac{3}{e^x}\right]_1^b = 3e^{-1} - \lim_{b\to\infty} \frac{3}{e^b} = 3e^{-1}.$$

$\Rightarrow$ $\int_1^{\infty} \frac{3}{e^x+5}\,dx$ converges.

Extra Examples:

1. $\int_0^{\ln 2} \frac{1}{x^2 e^{1/x}}\, dx$ (Integrate directly)

2. $\int_{\pi}^{\infty} \frac{2+\cos x}{x}\, dx$ (Direct Comparison)

3. $\int_1^{\infty} \frac{1}{\sqrt{e^x - 1}}\, dx$ (Limit Comparison)

Why LCT works?

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L, \quad 0 < L < \infty.$$

Suppose $\int_a^\infty g(x)\,dx$ converges. We want to show that $\int_a^\infty f(x)\,dx$ also converges.

Sketch Proof: By def^n of limits, there exists N (large) such that

$$L-1 < \frac{f(x)}{g(x)} < L+1 \quad \text{for all } x > N.$$

For $x > N$, $f(x) < (L+1)g(x)$.

bounded finite — B

$$\int_a^\infty f(x)\,dx = \int_a^N f(x)\,dx + \int_N^\infty f(x)\,dx \le \boxed{\int_a^N f(x)\,dx} + \boxed{\int_N^\infty (L+1)g(x)\,dx}.$$

$$B = (L+1)\int_N^\infty g(x)\,dx \le (L+1)\underbrace{\int_a^\infty g(x)\,dx}_{\text{converges}} \Rightarrow \text{converges}.$$

$$\Rightarrow \int_a^\infty f(x)\,dx \text{ converges}. \quad \square$$